

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Let A and B be two sets. If A and B are disjoint then:

1 / 1 point

- ☒ $A - B = A$
- ☐ $A \cup B = \emptyset$
- ☐ $A \oplus B = \emptyset$
- ☐ $A = B$

👍 Nice work

$A \cap B = \emptyset \implies A - B = A$

2. Given a set $A = \{0, 1, 2, 5\}$ and a set $B = \{0, 1, 5, 7\}$, which one of the following sets represents $A \cap B$?

1 / 1 point

- ☐ $\{7\}$
- ☐ $\emptyset$
- ☐ $\{0, 1, 2, 5, 7\}$
- ☒ $\{0, 1, 5\}$

👍 Nice work

0, 1 and 5 are the only elements in both sets A and B , hence $A \cap B = \{0, 1, 5\}$

3. Given a set $A = \{1, 2, 4, 5, 7, 9\}$ and $B = \{8, 9, 4, 5, 1\}$, which one of the following sets represents $A \cup B$?

1 / 1 point

- ☒ $\{1, 2, 4, 5, 7, 8, 9\}$
- ☐ $\{2, 7, 8\}$
- ☐ $\{2, 7\}$
- ☐ $\{1, 4, 5, 9\}$

👍 Nice work

The union of A and B contains the elements in A or in B . In this case $A \cup B = \{1, 2, 4, 5, 7, 8, 9\}$

4. Given a set $A = \{a, b, c\}$ and $B = \{d, e, g\}$, which one of the following represents $A - B$?

1 / 1 point

- ☐ $\{b, e, g\}$
- ☐ $\{a, b, c, d, e, g\}$
- ☐ $\emptyset$
- ☒ $\{a, b, c\}$

👍 Nice work

$A - B$ consists of just the elements in A which are not in B . In this case $A - B = \{a, b, c\}$

5. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3\}$. Which one of the following represents the set $A \oplus B$?

1 / 1 point

- ☐ $\{1, 2, 3, 3, 2, 1\}$
- ☐ $\{1, 2, 3\}$
- ☒ $\emptyset$
- ☐ $\{1, 2, 3, 1, 2, 3\}$

👍 Nice work

$A \oplus B$ consists of just the elements in A or in B , but not in both. $A \oplus B = A \cup B - (A \cap B)$

6. Let A and B be two sets. Given the following membership table:

1 / 1 point

A	B	S
0	0	0
0	1	1
1	0	1
1	1	1

Which one of the following sets represents the set S ?

- ☐ $A \cap B$
- ☒ $A \cup B$
- ☐ $A - B$
- ☐ $A \oplus B$

👍 Nice work

$A \cup B$ contains element that are in A or in B

7. Let A, B, C and S be four sets. Given the following membership table:

1 / 1 point

A	B	C	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Which one of the following sets represents the set S ?

- ☐ $(A \cap B) \cup C$
- ☐ $A - B - C$
- ☐ $A \oplus B \oplus C$
- ☒ $(A \cup B) \cap C$

👍 Nice work

This set contains elements in C which are in A or in B